AN INTELLIGENT ASSISTANT FOR FERTILITY PREDICTION, WELLNESS, AND SUPPORT

Project ID – 25-26J-246

Project Proposal Report

Bachelor of Science (Hons) Degree in Information Technology
Specializing in

Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka (SLIIT)

August 2025

AN INTELLIGENT ASSISTANT FOR FERTILITY PREDICTION, WELLNESS, AND SUPPORT (IVF SMART WATCH)

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August-2025

DECLARATION

I hereby declare that the work presented in this report of proposal is my own and has not been submitted elsewhere. All sources of information have been duly acknowledged. This report is submitted in partial fulfillment of the requirements of the above degree program, and it reflects the research and development plan for my individual component of the project.

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The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

…….

……..…… Signature of the Supervisor

Date

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……...… Signature of the Co-Supervisor

Date

Mrs. Vishaliney Pirathap

ABSTRACT

Infertility is a growing health issue affecting many couples worldwide. In-Vitro Fertilization (IVF) is one of the most effective treatments, but its success rate is often limited due to lack of continuous monitoring and timely decision-making. Traditional IVF monitoring methods depend on periodic hospital visits, which may miss important changes in a patient's condition. This research proposes a wearable IoT device designed specifically for IVF patients to monitor key physiological parameters in real time and provide personalized guidance.

The proposed device will integrate sensors for body temperature, spo2 levels, heart rate variability, and stress. Data collected will be processed by a microcontroller and transmitted securely to a cloud platform. Artificial Intelligence (AI) and Machine Learning (ML) models will analyze the data to identify fertility stages, predict IVF progress. To ensure data security and privacy, techniques such as federated learning and blockchain will be implemented.

This project aims to bridge the gap between existing IVF monitoring methods and the need for real-time, personalized care. The device is designed to be comfortable, accurate, and affordable, making it suitable for long-term use during the IVF process. Clinical validation and usability testing will be conducted to ensure the reliability of both the hardware and software components.

By combining wearable IoT technology, AI analytics, and secure data management, this research hopes to improve IVF treatment outcomes, reduce patient stress, and enhance the overall fertility care experience.

Keywords

IVF, Infertility, Wearable IoT, AI, Machine Learning, Fertility Tracking, Real-time Monitoring, Secure Data.

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LIST OF ABBREVIATIONS

AI – Artificial Intelligence

AI/ML – Artificial Intelligence / Machine Learning

ART – Assisted Reproductive Technologies

BT – Body Temperature

CS – Cloud Storage

DM – Data Management

DPS – Data Privacy and Security

EBD – Evidence-Based Decisions

HR – Heart Rate

HRV – Heart Rate Variability

IoT – Internet of Things

IVF – In-Vitro Fertilization

NI – Non-Invasive

RT – Real-Time

HIPAA - Health Insurance Portability and Accountability Act

GDPR - General Data Protection Regulation

1. INTRODUCTION

In recent years, infertility has become a significant health concern, affecting millions of couples worldwide. For many families, the inability to conceive naturally leads to emotional stress, social pressure, and even psychological challenges. Among the available medical solutions, In-Vitro Fertilization (IVF) has emerged as one of the most widely used Assisted Reproductive Technologies (ART). IVF has given hope to many couples, yet it is a complex medical process that requires careful planning, close monitoring, and strong emotional support. The success of IVF largely depends on tracking and managing a variety of physiological and lifestyle factors such as hormonal balance, body temperature (BT), heart rate (HR), stress level, and sleep quality throughout different stages of the treatment.

Traditionally, IVF treatment monitoring has relied heavily on frequent hospital visits, manual tracking of health records, and periodic testing. While these methods provide essential information, they often fail to capture real-time (RT) changes in a patient's condition. For instance, stress levels or sleep disturbances that occur at home may go unnoticed during hospital checkups, even though these factors have a direct impact on treatment outcomes. Furthermore, the repeated hospital visits can be physically and emotionally exhausting for patients, adding more stress to an already sensitive situation.

The rapid growth of Internet of Things (IoT) devices has introduced new possibilities for improving IVF care. Wearable devices, such as smartwatches and fitness trackers, are capable of continuously monitoring key health parameters in a non-invasive (NI) way. They can measure vital signals include Heart Rate Variability (HRV), BT, sleep cycles, and stress levels, and transfer this data to secure cloud storage (CS) in RT. This allows patients and healthcare providers to have a continuous, accurate picture of health trends instead of relying only on periodic checkups.

The integration of Artificial Intelligence (AI) into healthcare adds another powerful layer to this process. AI/ML (Artificial Intelligence / Machine Learning) models can analyze vast amounts of data collected by wearables, identify subtle patterns, and provide personalized recommendations. For IVF patients, this means more precise

guidance, such as optimized diet suggestions, stress management tips, and predictions of treatment success probability. These AI-driven insights can empower patients to take proactive measures.

Despite the availability of general-purpose wearable devices, there is still a noticeable research and application gap. Most existing devices are not specifically designed to support the unique needs of IVF patients. They may track basic fitness parameters but lack the ability to integrate critical medical, behavioral, and psychological data relevant to IVF. Additionally, current solutions often do not address data privacy and security (DPS) concerns adequately, which is crucial given the sensitive nature of reproductive health data.

This gap highlights the need for a specialized wearable and AI-driven system that combines continuous physiological monitoring, emotional and behavioral support, secure data management (DM), and personalized treatment recommendations. By bridging this gap, the proposed system aims to improve the success rates of IVF, reduce patient stress, and provide couples with holistic support throughout their fertility journey. Ultimately, this approach not only enhances medical outcomes but also promotes the emotional well-being of patients navigating the challenges of infertility treatment.

1.1 BACKGORUND AND LITERATURE SURVEY

1.1.1 IVF Watch – Smart Wearable Device

Infertility management and assisted reproductive technologies such as IVF require continuous health monitoring and personalized decision support. Recent studies

highlight the role of wearable and IoT-based devices in improving patient outcomes. For example, biosensor technologies have been applied in healthcare for real-time monitoring of physiological signals, demonstrating potential for fertility applications [1], [2]. Wearable devices integrated with AI and machine learning further enhance prediction accuracy and provide individualized health recommendations [3], [4].

Several works have emphasized the value of IoT frameworks in remote health monitoring. Studies on smart healthcare systems indicate that cloud-based storage, wireless data transfer, and secure communication protocols are essential for sensitive applications include IVF [5]. Additionally, AI-driven solutions such as deep learning and neural networks have been successfully implemented in fertility and women's health applications, improving diagnosis and treatment guidance[6], [7].

However, existing wearable solutions mainly focus on general fitness or broad healthcare domains rather than IVF-specific monitoring. Research also highlights limitations in validation, patient usability, and privacy concerns[8]. Given the sensitive nature of reproductive health data, incorporating blockchain and federated learning has been suggested to ensure transparency and secure handling of patient information.

Therefore, there remains a significant gap in developing a specialized wearable system tailored to IVF patients that integrates continuous physiological monitoring, emotional support, secure data management, and AI-driven recommendations. Bridging this gap could enhance treatment success rates and reduce patient stress.

1.2 Research Gap

Infertility treatment and assisted reproductive technologies such as IVF require continuous health monitoring and personalized decision-making support. Existing research highlights the potential of wearable sensors, IoT devices, and AI-driven healthcare platforms for real-time tracking of physiological parameters and improved patient outcomes[1, 9]. Studies also emphasize the importance of secure cloud

platforms, wireless communication, and privacy-preserving techniques such as blockchain and federated learning for handling sensitive health information [5 ,2]. Wearable devices have also been applied in areas such as chronic disease management, stress detection, and remote patient monitoring, demonstrating strong potential for clinical use [7] .

However, most current research focuses on general healthcare or fitness monitoring rather than IVF-specific applications. Existing solutions often fail to integrate comprehensive physiological monitoring, such as spo2 levels, body temperature, heart rate variability, and stress, with AI-based personalized fertility insights. Moreover, gaps remain in validating the usability, clinical accuracy, and reliability of wearable systems in IVF contexts. Privacy and trust issues also persist, as reproductive health data require robust frameworks for secure storage and transparent sharing [10].

Therefore, there is a clear research gap in developing a specialized, secure, and AI-powered wearable IoT solution tailored for IVF patients, aiming to enhance treatment outcomes and reduce emotional stress.

1.3 Research Problem

Despite advances in fertility treatments include IVF, many patients face challenges due to the lack of continuous, real-time monitoring of critical physiological parameters throughout the treatment stages. Current methods often depend on periodic clinical visits and manual data collection, which may miss important changes affecting treatment success. Additionally, existing wearable health devices are generally designed for general health monitoring and lack tailored features specific to IVF needs.

Moreover, the sensitive nature of IVF-related data raises concerns about privacy and security, which are not adequately addressed in most wearable systems. There is also

a need for intelligent systems that can provide stage-specific, personalized decision support to patients and healthcare providers.

Therefore, the research problem is how to develop a wearable IoT device that enables continuous, accurate, and secure monitoring of fertility-related parameters, integrates artificial intelligence for real-time guidance during IVF treatment, and ensures patient data privacy throughout the process.

2. OBJECTIVES

2.1 Main Objective

Design and Development of Wearable IoT Device:

Create a comfortable, user-friendly wearable device specifically tailored for IVF patients that can continuously monitor vital physiological parameters relevant to fertility and IVF treatment stages.

Integration of Advanced Multi-Sensor Technologies:

Incorporate a variety of sensors such as spo2 level detectors, body temperature sensors, photoplethysmography (PPG) for heart rate and variability, accelerometers for activity tracking, and stress-related sensors to capture comprehensive reproductive health data.

Implementation of AI-Based Data Analytics:

Develop and integrate artificial intelligence and machine learning algorithms capable of processing real-time sensor data, detecting patterns, predicting critical fertility windows, and providing personalized, stage-specific decision support to both patients and healthcare providers.

Ensuring Data Privacy and Security:

Implement robust data protection mechanisms, including federated learning and blockchain technology, to secure sensitive fertility and IVF data, ensuring patient confidentiality and compliance with privacy regulations.

Evaluation of System Performance and User Experience:

Conduct thorough testing of the wearable device for accuracy, reliability, comfort, and usability, along with clinical validation to assess its impact on improving IVF treatment outcomes and patient satisfaction.

2.2 Specific Objectives

Identify and select key physiological parameters crucial for IVF monitoring to be measured by the wearable device.

Design the hardware architecture of a wearable IoT device incorporating sensors for spo2 levels, temperature, heart rate variability, and stress.

Develop software algorithms for real-time data acquisition, processing, and transmission from the wearable device to a secure cloud platform.

Create AI and machine learning models that analyze sensor data and provide personalized fertility and IVF stage-specific recommendations.

Implement privacy-preserving techniques such as federated learning and blockchain for secure data storage and sharing.

Conduct usability testing and validate the accuracy and reliability of the device in clinical or simulated IVF environments.

Assess patient satisfaction and the impact of the wearable device on improving IVF treatment outcomes

3. METHODOLOGY

This research will follow a systematic approach to design, develop, and evaluate a wearable IoT device for IVF monitoring. The methodology includes several key phases: requirement analysis, device design, AI integration, data security implementation, and evaluation.

Requirement Analysis:

The first step involves identifying the critical physiological parameters needed for effective IVF monitoring. This will be achieved through literature review and consultations with fertility experts and clinicians to understand the key health indicators, such as hormonal levels, body temperature, heart rate variability, and stress markers.

Device Design and Development:

Based on the requirements, the hardware architecture of the wearable device will be designed. This includes selecting appropriate sensors (e.g., spo2 sensors, temperature sensors, photoplethysmography sensors, accelerometers) and microcontrollers capable of continuous data acquisition. The device will be designed to be ergonomic and comfortable for long-term use by patients. Firmware development will focus on efficient real-time data collection, preprocessing, and wireless transmission to a cloud server using protocols include Bluetooth Low Energy (BLE).

AI and Machine Learning Integration:

Collected sensor data will be sent to a cloud-based platform where AI and machine learning models will analyze the data. These models will identify patterns related to fertility stages and predict optimal intervention points. Algorithms will be trained using anonymized datasets and validated for accuracy in fertility prediction and IVF readiness assessment. Personalized feedback and alerts will be generated to assist patients and clinicians.

Data Privacy and Security:

Given the sensitivity of IVF data, privacy-preserving techniques will be implemented. Federated learning will allow AI models to train on decentralized data without sharing raw patient information, while blockchain technology will secure data storage and access logs, ensuring transparency and trust.

Evaluation and Validation:

The device will undergorigorous testing for sensor accuracy, data transmission reliability, battery life, and user comfort. Clinical validation will be conducted with volunteers undergoing IVF to assess the effectiveness and usability of the system. Feedback will be gathered to refine the design and algorithms.

This methodology aims to develop a comprehensive, secure, and user-friendly wearable solution to support IVF treatment.

4. GANTT CHART

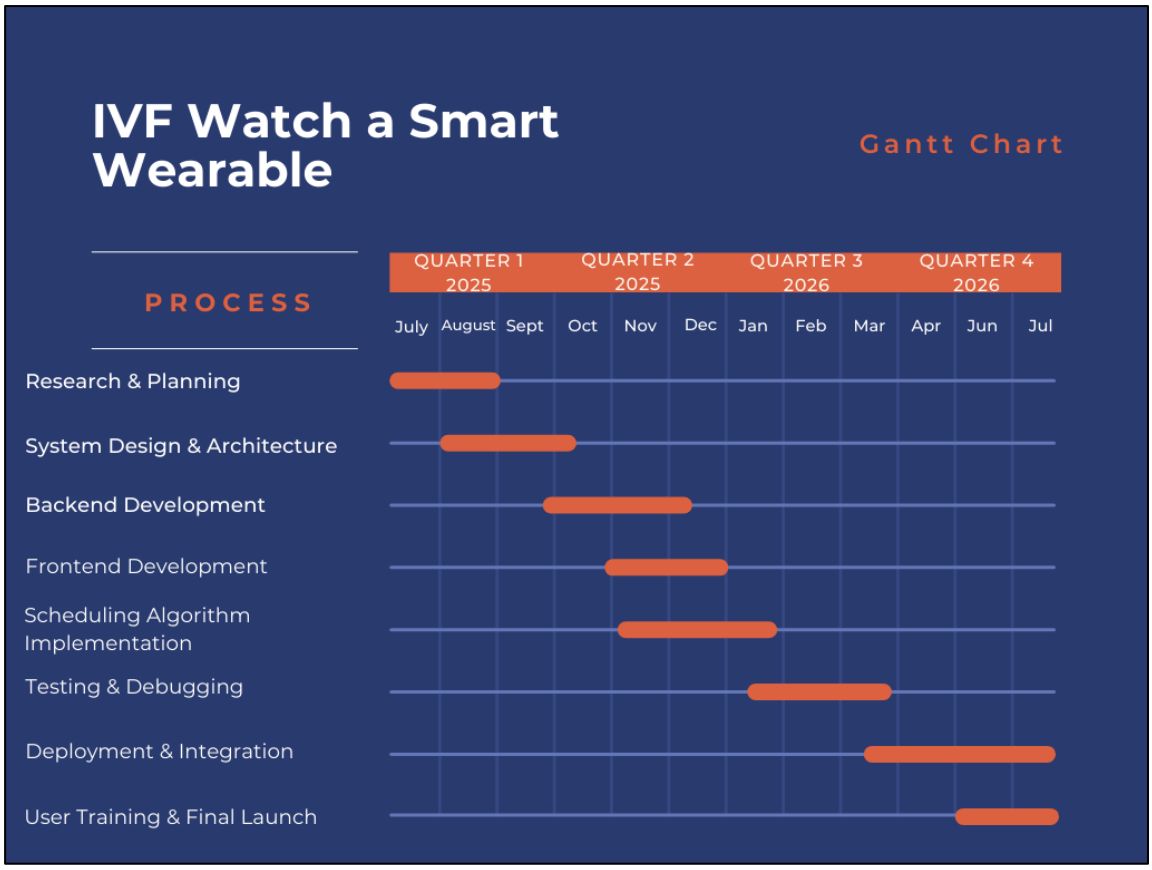

Figure 4.1 Gantt Chart

5. PROJECT REQUIREMENTS

5.1 Functional Requirements

Continuous Physiological Data Monitoring

The device shall continuously monitor key physiological parameters such as spo2 levels, body temperature, heart rate variability, and stress indicators.

Real-Time Data Transmission

The wearable shall transmit collected data in real-time to a cloud-based platform via a secure wireless protocol include Bluetooth Low Energy (BLE).

AI-Based Data Analysis

The system shall analyze incoming data using AI and machine learning models to detect fertility stages, predict IVF readiness, and identify critical intervention points.

Personalized Feedback and Alerts

Based on the analysis, the system shall provide personalized recommendations, alerts, and notifications to patients and healthcare providers.

Secure Data Storage and Privacy

The platform shall securely store patient data with encryption, employing privacy-preserving technologies such as federated learning and blockchain to protect sensitive information.

User Interface Integration

The device shall be integrated with a mobile or web application to allow patients and clinicians to view real-time data, historical trends, and receive AI-generated advice.

Battery and Power Management

The wearable shall manage power efficiently to support continuous monitoring for a minimum predefined duration (e.g., 24-48 hours) without frequent recharging.

Sensor Calibration and Accuracy

The device shall regularly calibrate sensors to maintain measurement accuracy within clinically acceptable ranges.

Data Synchronization and Backup

The system shall synchronize data between the device and cloud server, ensuring no loss of information and enabling data backup.

5.2 Non-Functional Requirements

Performance

The system shall process and analyze physiological data with minimal delay to provide timely feedback and alerts.

Reliability

The wearable device and associated software shall operate continuously without failures for extended periods during the IVF treatment cycle.

Usability

The device and user interface shall be intuitive and easy to use for patients and healthcare providers with minimal training.

Scalability

The cloud platform shall be scalable to support increasing numbers of users and data volumes without degradation in performance.

Security

The system shall implement strong security measures to prevent unauthorized access, including encrypted communication, authentication, and secure data storage.

Privacy Compliance

The system shall comply with relevant healthcare data privacy regulations (such as HIPAA or GDPR) to protect patient information.

Maintainability

The hardware and software shall be designed for easy maintenance, updates, and upgrades without significant downtime.

Portability

The wearable device shall be lightweight and compact to allow easy wearing during daily activities.

Battery Life

The device shall support long battery life, ideally lasting 24 to 48 hours on a single charge to minimize interruptions.

Interoperability

The system shall be compatible with various mobile devices and healthcare information systems to allow seamless data exchange.

6. DIAGRAMS

6.1 SYSTEM DIAGRAM

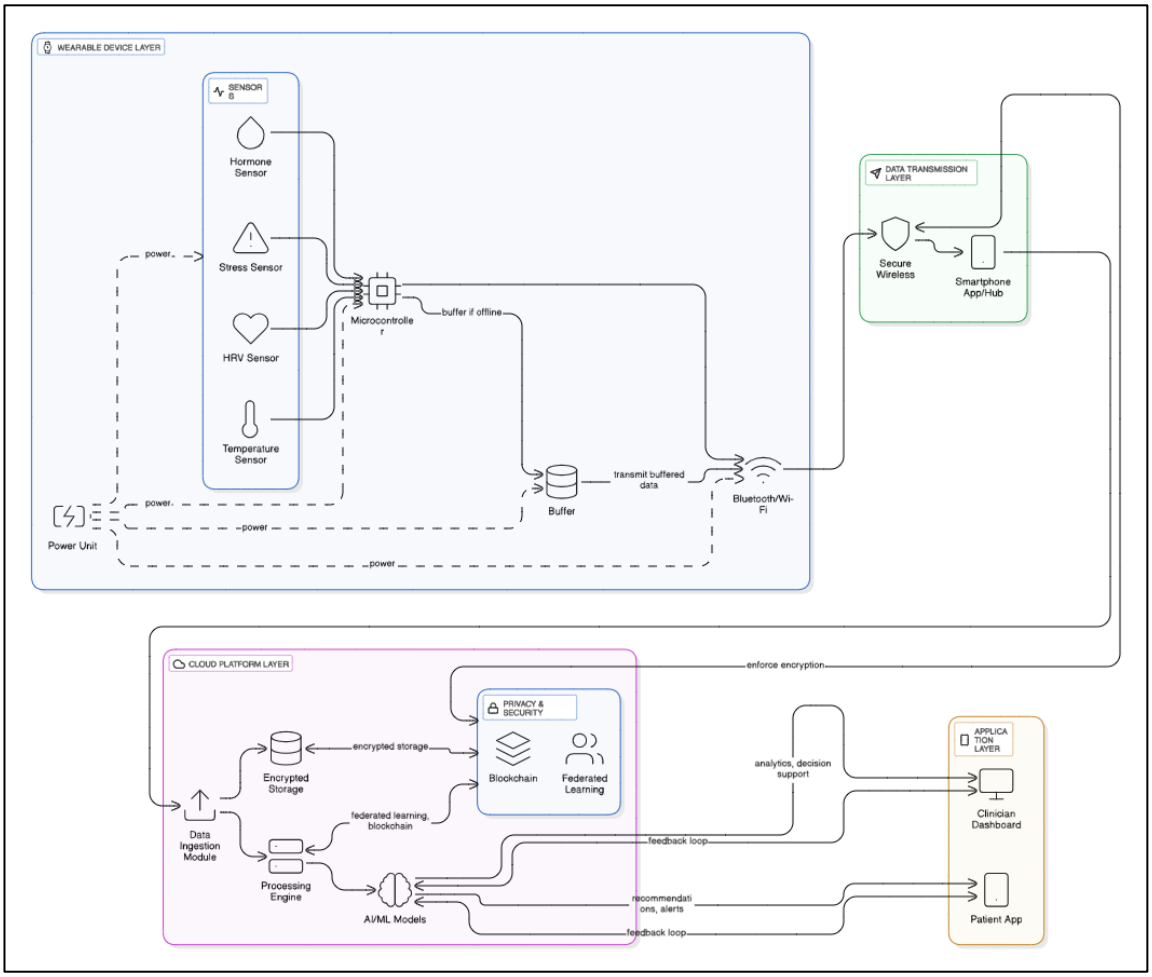

Figure 6.1 System Diagram

6.2 WORKFLOW DIAGRAM

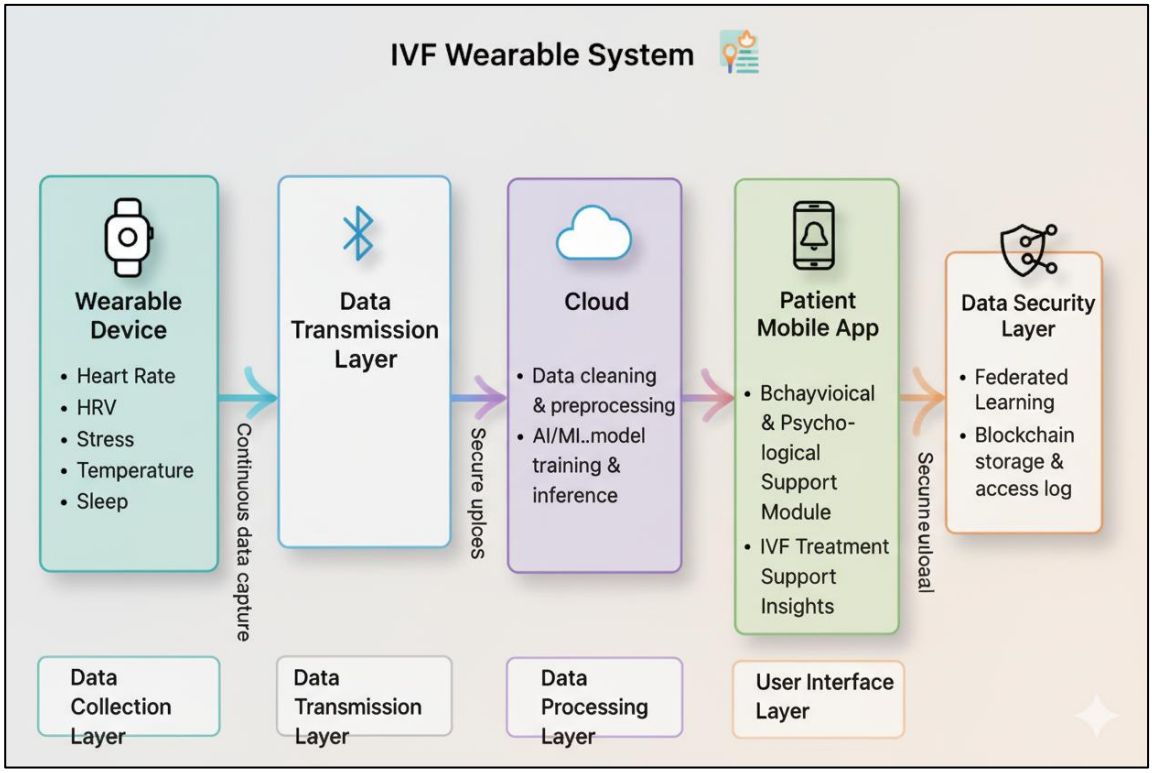

Figure6.2 Workflow Diagram

7. CONCLUSION

Wearable IoT technology offers a promising avenue to enhance fertility and IVF treatment by enabling continuous, personalized monitoring of key physiological parameters. Traditional IVF monitoring methods, which rely heavily on periodic clinical visits and manual data collection, often fail to capture critical real-time changes necessary for improving treatment success rates. The integration of advanced sensors with artificial intelligence and machine learning models can provide deep insights into fertility stages, allowing for stage-specific guidance and timely interventions tailored to individual patients. This research aims to bridge existing gaps by developing a dedicated IVF wearable device that combines multi-sensor data acquisition with intelligent analytics and robust privacy protections. By incorporating spo2 sensors, temperature sensors, heart rate variability monitoring, and stress indicators, the device will provide comprehensive and accurate physiological data. AI-driven models will analyze this data in real time to offer personalized feedback and predictive recommendations, supporting patients and healthcare providers throughout the IVF journey.

Data privacy and security remain paramount, especially given the sensitive nature of fertility data. This project will leverage cutting-edge privacy-preserving techniques, such as federated learning and blockchain technology, to ensure secure data storage and sharing without compromising patient confidentiality. Through rigorous design, development, and clinical validation, this research seeks to produce a reliable, user-friendly, and secure wearable system that can improve IVF outcomes and patient experiences. The successful implementation of such technology holds the potential to transform fertility treatment by making it more adaptive, informed, and patient-centered.

8. BUDGET AND BUDGET JUSTIFICATION

Table 8.1 Custom IVF Watch Build Budget – Hardware Components

Item	Qty	Cost per Unit (LKR)	Total (LKR)	Justification
ESP32-based Dev Board (LilyGo T-Wristband)	1	6600.00	6600.00	Core microcontroller with BLE for data processing & communication
PPG Sensor (MAX30102 module)	1	1650.00	1650.00	Measures heart rate & HRV for stress detection
GSR Sensor Module	1	2640.00	2640.00	Detects stress via skin conductance
Temperature Sensor (MAX30205)	1	1980.00	1980.00	Tracks body temperature (important for IVF cycle monitoring)
Accelerometer (MPU6050)	1	990.00	990.00	Used for sleep & activity pattern detection
Vibration Motor + Driver	1	660.00	660.00	Provides medication & water reminders through vibration alerts
Li-Po Battery (300 mAh)	1	1320.00	1320.00	Power source for continuous operation
Wrist Strap + Enclosure	1	1650.00	1650.00	Wearable design for comfort & usability
Miscellaneous (wires, PCB, connectors)		2310.00	2310.00	Additional components for assembly & circuit connections
Charging Module (TP4056)	1	660.00	660.00	Charging capability for battery
Total			20,460.00	Cost-effective, complete control over features

Table 8.2Custom IVF Watch Build Budget – Software Components

Item	Qty	Cost per Unit (LKR)	Total (LKR)	Justification
Arduino IDE + Libraries (Open Source)	-	0	0	Free
Firmware Development (ESP32)	-	5,000	5,000	Custom coding effort
Mobile App Development (Flutter SDK)	-	0	0	Free SDK (open source)
Testing Devices (Android phone for app test)	1	35,000	35,000	For Bluetooth & app testing
Testing Equipment (Multimeter, soldering kit)	-	5,000	5,000	For assembly & debugging
Cloud Hosting (Firebase for app backend)	-	3,300/month × 2 months = 6,600	6,600	Data sync, reminders
Software Sub-Total			51,600	
Grand Total (Hardware + Software)			72,060	Complete IVF Watch build

Table 8.3 Existing Smartwatch Customization Budget

Item	Qty	Cost per Unit (LKR)	Total (LKR)	Justification
Wear OS Smartwatch (TicWatch / Fossil Gen 6)	1	49500.00	49500.00	Ready-made smartwatch with HR & motion sensors, runs Wear OS for app integration
Developer Account (Google Play)	1	8250.00	8250.00	Required for publishing companion app & watch app
Misc Testing Accessories		3300.00	3300.00	Extra straps, charging docks, BLE testing tools
Total			61050.00	Less development effort, but higher cost and limited customization

9. COMMERCIALIZATION

The proposed IVF wearable monitoring module has strong commercialization potential due to the growing demand for personalized fertility management solutions.

Globally, infertility affects millions of couples, and the IVF market is projected to grow steadily, creating opportunities for innovative digital health devices. The module integrates physiological monitoring (spo2 levels, temperature, HRV, and stress), AI-driven fertility recommendations, and secure IoT connectivity, offering a unique value proposition compared to existing general-purpose wearables.

Target Market: Fertility clinics, IVF centers, gynecology hospitals, and individual patients undergoing assisted reproductive treatments.

Revenue Model: The device can be commercialized through a hybrid model — one-time device purchase combined with subscription-based access to advanced AI analytics, cloud storage, and personalized recommendations. Partnerships with IVF clinics can ensure wider adoption and integration into treatment packages.

Competitive Advantage: Unincluded fitness wearables, this solution is specialized for reproductive health, integrating secure data handling (blockchain, federated learning) and clinical-grade accuracy validated in healthcare environments.

Scalability: Beyond IVF, the module can be adapted for broader women's health monitoring, stress management, and chronic condition tracking, expanding its commercial scope.

By addressing an unmet need in IVF treatment and ensuring regulatory compliance, the module has potential to become a leading fertility-focused digital health solution.

REFERENCES

[1] N. Almtireen, H. Altaha, A. Alissa, M. Ryalat, and H. Elmoaqet, "AI-Driven Mobile App for Personalized Health Monitoring," in *2024 22nd International Conference on Research and Education in Mechatronics, REM 2024*, Institute of Electrical and Electronics Engineers Inc., 2024, pp. 338–342. doi: 10.1109/REM63063.2024.10735595.

[2] E. G. Lowery-Gionta *et al.*, “Initial development of the Stress Monitoring and Response Tool (SMART): A holistic measure of stress following trauma,” *PLoS One*, vol. 20, no. 6 June, Jun. 2025, doi: 10.1371/journal.pone.0321939.

[3] *2020 IEEE 5th International Symposium on Smart and Wireless Systems within the Conferences on Intelligent Data Acquisition and Advanced Computing Systems (IDAACS-SWS)*. IEEE, 2020.

[4] T. Inui *et al.*, “Use of a Smart Watch for Early Detection of Paroxysmal Atrial Fibrillation: Validation Study,” *JMIR Cardio*, vol. 4, no. 1, Jan. 2020, doi: 10.2196/14857.

[5] Y. Gu, J. Shen, and Y. Chen, “Poster Abstract: Know You Better: A Smart Watch Based Health Monitoring System,” in *Proceedings - 4th IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technologies, CHASE 2019*, Institute of Electrical and Electronics Engineers Inc., Sep. 2019, pp. 7–8. doi: 10.1109/CHASE48038.2019.00009.

[6] G. Ji, C. Msigwa, D. Bernard, G. Lee, J. Woo, and J. Yun, “Health24: Health-related Data Collection from Wearable and Mobile Devices in Everyday Lives,” in *Proceedings - 2023 IEEE International Conference on Big Data and Smart Computing, BigComp 2023*, Institute of Electrical and Electronics Engineers Inc., 2023, pp. 336–337. doi: 10.1109/BigComp57234.2023.00074.

[7] N. Kallivalappil, K. D’Souza, A. Deshmukh, C. Kadam, and N. Sharma, “Empath.ai: a Context-Aware Chatbot for Emotional Detection and Support,” in *2023 14th International Conference on Computing Communication and Networking Technologies, ICCCNT 2023*, Institute of Electrical and Electronics Engineers Inc., 2023. doi: 10.1109/ICCCNT56998.2023.10306584.

[8] I. Moraiti and A. Drigas, “Measuring the stress of autistic people with the help of a smartwatch, Internet of Things Technology,” *Brazilian Journal of Science*, vol. 3, no. 2, pp. 45–56, Feb. 2024, doi: 10.14295/bjs.v3i2.466.

[9] G. V. Heringer and D. R. Vinson, “Utility of smart watch in expediting diagnosis of cold drink-triggered atrial fibrillation: a case report,” *Int J Emerg Med*, vol. 17, no. 1, Dec. 2024, doi: 10.1186/s12245-024-00716-z.

[10] Ü. T. Erbay *et al.*, “Are Smart Watches Really Smart? Comparison of Blood Oxygen Saturation Values Measured by Smart Watch, Pulse Oximetry and Arterial Blood Gases in Patients with Chronic Obstructive Pulmonary Diseases,” *International Journal of COPD*, vol. 20, pp. 1457–1463, 2025, doi: 10.2147/COPD.S500643.